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Original Research Article

Survival in patients with neuroendocrine tumors of the colon, rectum and small intestine

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A B S T R A C T

Background: Neuroendocrine neoplasms (NENs) of the colon, rectum and small intestine (SI) are increasing in incidence and prevalence. We evaluated the 5-year overall survival (OS), and cancer-specific survival (CSS).

Methods: The Surveillance, Epidemiology, and End Results (SEER) 18 registry from 2000 to 2017 was accessed to identify patients with colonic, rectal, and SI NENs.

Results: 46,665 patients were diagnosed with NENs of the colon (n = 10,518, 22.5%), rectum (18,063, 38.7%), and SI (18,084, 38.8%). By tumor site alone, patients with well-differentiated neuroendocrine tumors (NETs) of the rectum had improved 5-year OS (HR 0.72, 95% CI 0.68–0.77, $p < 0.001$). However, patients with rectal poorly-differentiated neuroendocrine carcinomas (NECs) who underwent oncologic resection had lower 5-year OS (35.1%) compared to colon (41.9%), and SI (72.5%).

Conclusions: Surgical resection may improve 5-year OS for NECs of the SI and colon, except in the rectum where survival was reduced. More frequent surveillance and timely initiation of systemic therapy should be considered for rectal NECs.

1. Introduction

Neuroendocrine neoplasms (NENs) of the colon, rectum and small intestine (SI), although rare, are increasing in incidence and prevalence.[1–6] Dasari et al. described an age-adjusted incidence among all NENs of any site increased by 6.4-fold from 1973 to 2012, based on SEER data.[5] Furthermore, the highest increase in incidence rates was found in the small intestine (0.8 cases per 100,000), rectum (0.65/100,000), and pancreas (0.33/100,000), as per SEER 1975 to 2012 data.[6] With regard to prevalence among all NENs, this also increased from 0.006% in 1993 to 0.048% in 2012.[5] Prevalence was noted to be highest within the rectum, then lung and SI.[5]

NENs within the gastrointestinal tract are more commonly located in the small intestine (38%), followed by rectum (34%), colon (16%), and stomach (11%).[7,8] However, the proportion of rectal NENs has now surpassed that of the small intestine due to widespread implementation of screening colonoscopy.[7,9–16] Prognosis has been predicted to be associated with the anatomical primary site of origin – either foregut, midgut, or hindgut, with some studies suggesting reduced survival with foregut or midgut primary NENs compared with hindgut.[16,17] Other studies suggest the symptoms at presentation and stage of disease at diagnosis as more predictive of prognosis.[16–23]

The WHO classification of neuroendocrine neoplasms has recently been updated in 2019 to include three grades of well-differentiated neuroendocrine tumors (NETs) (including low, intermediate, and high), poorly differentiated neuroendocrine carcinomas (NECs) divided into small cell type and large cell type, and mixed neuroendocrine non-neuroendocrine neoplasms (MiNENs).[24,25] Previously published studies report a 5-year OS amongst all midgut neuroendocrine neoplasms as 79% for low-grade, 74% for intermediate-grade, and 40% for high-grade tumors.[4,5,26] Another study reported colonic NENs with worse prognosis compared to those located in the rectum.[4] Interestingly, reported overall survival rates for all neuroendocrine neoplasms, irrespective of tumor site, improved from 2000 to 2004 compared to 2009–2012, including advanced gastrointestinal NENs.[5] This finding may reflect improvements both in detection of early-stage NENs (including octreoscans and DOTATATE imaging), and in treatment modalities for advanced NENs (such as the RADIANT-4 trial with everolimus, and the NETTER-1 trial with octreotide and DOTATATE analogs).[5,26–28] While most NENs are indolent, well-differentiated NETs with an associated improved prognosis, at least 10–20% of NENs are poorly-differentiated and rapidly progressive NECs.[6,26] Whether improvements in early detection or treatment of advanced NENs are associated with differences in overall survival between tumor site, histopathological subtype or grade remain unclear. We thus aimed to directly compare the 5-year OS and CSS between different tumor types among colonic, rectal, and SI NENs using

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Table 1
Patient demographics and tumor type.

	Colon (n = 10518)	Rectum (n = 18063)	Small Intestine (n = 18084)	*p* value
Age, years				
<49	3192 (30.9%)	3560 (20.1%)	2140 (12.0%)	<0.001
50–59	2487 (24.0%)	7710 (43.6%)	4071 (22.9%)	
60–69	2181 (21.1%)	4080 (23.1%)	5209 (29.3%)	
70–79	1567 (15.2%)	1787 (10.1%)	4160 (23.4%)	
>80	916 (8.9%)	546 (3.1%)	2221 (12.5%)	
Gender				
Female	5722 (54.4%)	8989 (49.8%)	8647 (47.8%)	< 0.001
Male	4796 (45.6%)	9074 (50.2%)	9437 (52.2%)	
Race				
American Indian/ Alaska Native	75 (0.7%)	165 (0.9%)	69 (0.4%)	< 0.001
Asian/Pacific Islander	429 (4.1%)	2454 (13.6%)	480 (2.7%)	
Black	1347 (12.8%)	4315 (23.9%)	3080 (17.0%)	
White	8480 (80.6%)	10125 (56.1%)	14284 (79.0%)	
Unknown	187 (1.8%)	1004 (5.6%)	171 (0.9%)	
WHO classification				
Well-differentiated NETs	6783 (64.5%)	15828 (87.6%)	14705 (81.3%)	< 0.001
Poorly differentiated NECs	2985 (28.4%)	2167 (12.0%)	3331 (18.4%)	
MiNENs	750 (7.1%)	68 (0.4%)	48 (0.3%)	
Grade				
I – well-differentiated	3968 (37.7)	5657 (31.3)	7753 (42.9)	< 0.001
II – moderately differentiated	900 (8.6)	954 (5.3)	1818 (10.1)	
III – poorly differentiated	1363 (13.0)	501 (2.8)	273 (1.5)	
IV – undifferentiated	562 (5.3)	292 (1.6)	81 (0.4)	
Unknown	3725 (35.4)	10659 (59.0)	8159 (45.1)	
Extent of resection				
Local	1529 (14.5%)	12949 (71.7%)	2566 (14.2%)	< 0.001
Oncologic	7398 (70.3%)	1178 (6.5%)	12288 (67.9%)	
None	1532 (14.6%)	3690 (20.4%)	3095 (17.1%)	
Unknown	59 (0.6%)	241 (1.3%)	135 (0.7%)	
Stage				
Localized	4898 (46.6%)	14438 (79.9%)	6445 (35.6%)	< 0.001
Regional	2632 (25.0%)	437 (2.4%)	6579 (36.4%)	
Distant	2379 (22.6%)	847 (4.7%)	3919 (21.7%)	
Unknown	609 (5.8%)	2341 (12.0%)	1141 (6.3%)	
Median survival, months (IQR)	27.0 (66.0)	67.0 (96.0)	52.0 (77.0)	< 0.001
Well-differentiated NETs	34.0 (77.0)	75.0 (96.0)	51.0 (83.0)	
Poorly differentiated NECs	11.0 (47.0)	21.0 (55.0)	52.0 (57.0)	
MiNENs	22.0 (43.8)	41.0 (109.8)	31.5 (99.0)	

WHO: World Health Organization. NETs: well-differentiated neuroendocrine tumors, NECs: poorly differentiated neuroendocrine carcinomas, MiNENs: mixed neuroendocrine non-neuroendocrine neoplasms. Surgery type none: not resected.

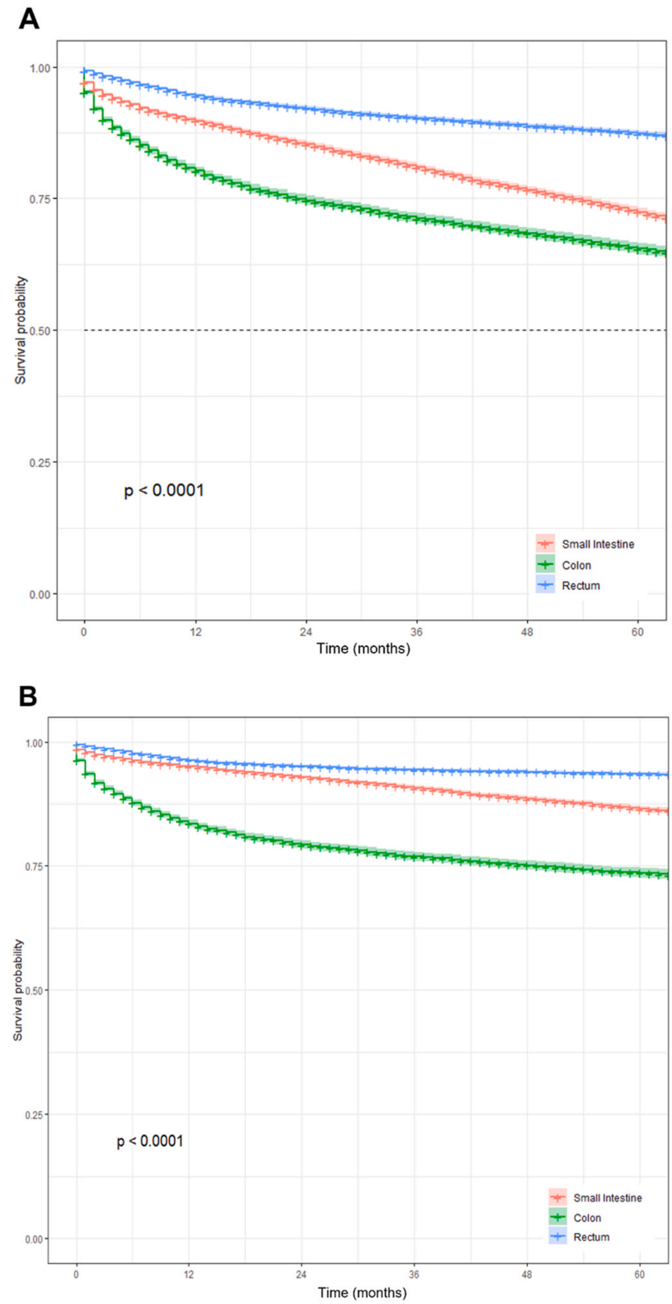

Fig. 1. Kaplan-Meier 5-year overall survival curves (**A**) and 5-year cancer-specific survival curves (**B**) by tumor site. Patient data collected using the SEER 18 registry from 2000 to 2017 among patients diagnosed with small intestinal (red), colonic (green), and rectal (blue) NENs. Abbreviations: Surveillance, Epidemiology, and End Results (SEER) database, neuroendocrine neoplasms (NENs). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

the SEER database. We expected to find improved 5-year OS and CSS in rectal NENs compared to colon and SI, as suggested in prior studies. Furthermore, we expected to find reduced 5-year OS and CSS in poorly-differentiated colonic NECs, especially those undergoing oncologic resection, as studies suggested a reduced 5-year OS in colonic NENs overall.[26,29,30]

Table 2
Univariate and multivariate hazard ratio survival analysis.

	Overall Survival				Cancer-specific Survival			
	Univariate HR (95% CI)		Multivariate HR (95% CI)		Univariate HR (95% CI)		Multivariate HR (95% CI)	
Age, years								
<49	–		–		–		–	
50–59	1.48 (1.37–1.60)	p < 0.001	1.51 (1.40–1.63)	p < 0.001	1.25 (1.14–1.38)	p < 0.001	1.28 (1.16–1.41)	p < 0.001
60–69	3.02 (2.80–3.25)	p < 0.001	2.64 (2.45–2.85)	p < 0.001	2.25 (2.05–2.47)	p < 0.001	1.79 (1.63–1.96)	p < 0.001
70–79	5.93 (5.51–6.38)	p < 0.001	4.68 (4.35–5.04)	p < 0.001	3.82 (3.47–4.19)	p < 0.001	2.63 (2.39–2.89)	p < 0.001
>80	12.27 (11.37–13.24)	p < 0.001	8.70 (8.05–9.40)	p < 0.001	7.20 (6.52–7.95)	p < 0.001	4.32 (3.91–4.78)	p < 0.001
Female	–		–		–		–	
Male	1.18 (1.14–1.22)	p < 0.001	1.21 (1.17–1.26)	p < 0.001	1.13 (1.08–1.19)	p < 0.001	1.16 (1.11–1.22)	p < 0.001
Tumor site								
Small Intestine	–		–		–		–	
Colon	1.20 (1.15–1.25)	p < 0.001	1.13 (1.08–1.19)	p < 0.001	1.91 (1.80–2.01)	p < 0.001	1.57 (1.48–1.68)	p < 0.001
Rectum	0.40 (0.38–0.42)	p < 0.001	0.72 (0.68–0.77)	p < 0.001	0.41 (0.38–0.43)	p < 0.001	1.16 (1.07–1.27)	p < 0.001
WHO classification								
NETs	–		–		–		–	
NECs	3.21 (3.09–3.33)	p < 0.001	1.59 (1.51–1.67)	p < 0.001	6.04 (5.74–6.36)	p < 0.001	2.04 (1.90–2.18)	p < 0.001
MiNENs	2.91 (2.62–3.23)	p < 0.001	1.32 (1.18–1.48)	p < 0.001	6.22 (5.54–6.98)	p < 0.001	1.65 (1.45–1.88)	p < 0.001
Stage								
Localized	–		–		–		–	
Regional	2.13 (2.03–2.23)	p < 0.001	1.30 (1.22–1.37)	p < 0.001	5.32 (4.90–5.78)	p < 0.001	3.05 (2.76–3.37)	p < 0.001
Distant	5.93 (5.69–6.19)	p < 0.001	3.04 (2.89–3.21)	p < 0.001	22.72 (21.09–24.248)	p < 0.001	10.12 (9.22–11.11)	p < 0.001
Unknown	1.45 (1.36–1.56)	p < 0.001	0.97 (0.90–1.05)	p = 0.495	2.45 (2.16–2.79)	p < 0.001	1.41 (1.23–1.62)	p < 0.001
Extent of resection								
Oncologic	–		–		–		–	
Local	0.34 (0.33–0.36)	p < 0.001	0.94 (0.88–1.00)	p = 0.067	0.12 (0.11–0.14)	p < 0.001	0.57 (0.51–0.65)	p < 0.001
None	1.40 (1.34–1.45)	p < 0.001	1.77 (1.69–1.86)	p < 0.001	1.51 (1.43–1.59)	p < 0.001	1.75 (1.64–1.87)	p < 0.001
Unknown	0.70 (0.57–0.85)	p < 0.001	1.66 (1.35–2.04)	p < 0.001	0.49 (0.35–0.67)	p < 0.001	1.64 (1.18–2.29)	p < 0.001

WHO: World Health Organization. NETs: well-differentiated neuroendocrine tumors, NECs: poorly differentiated neuroendocrine carcinomas, MiNENs: mixed neuroendocrine non-neuroendocrine neoplasms. Surgery type none: not resected.

2. Methods

2.1. Patients

The Surveillance, Epidemiology, and End Results (SEER) 18 registry from 2000 to 2017 was accessed to identify 46,665 patients diagnosed with colonic, rectal, and SI primary NENs. The SEER database collects data from cancer registries comprising approximately 34.6% of the US population.[31] The patients were selected from International Classification of Diseases (ICD) O-3 codes based on the 2008 WHO classification system, and subsequently stratified into the 2019 WHO classification of tumor grading: well-differentiated tumors subclassified into low grade (NET G1, mitotic rate <2 mitoses/2 mm^2, Ki67 index <3%), intermediate grade (NET G2, mitotic rate 2 to 20 mitoses/2 mm,[2] Ki67 index 3–20%), high grade (NET G3, mitotic rate >20 mitoses/2 mm,[2] Ki67 index >20%); poorly differentiated high grade tumors (neuroendocrine carcinoma or NECs) were subclassified as small cell type (SCNEC) or large cell type (LCNEC) where both have >20 mitotic rate and >20% Ki67 index; mixed neuroendocrine-non-neuroendocrine neoplasms (MiNENs) were either well or poorly differentiated with variable grading, mitotic rate and Ki67 index.[32] Those with well differentiated NETs (ICD codes: 8240/3, 8249/3, 8152/3, 8153/3, 8156/3, 8693/1, 8693/3, 8241/3, 8242/3), poorly differentiated NECs (8246/3, 8013/3, 8041/3), or mixed neuroendocrine non-neuroendocrine neoplasms (MiNENs) (8244/3) were thus included. The patients were further sub-classified by primary site of diagnosis, colon (ICD codes: cecum – C180, appendix – C181, ascending colon – C182, hepatic flexure – C183, transverse colon – C184, splenic flexure – C185, descending colon – C186, sigmoid colon – C187, large intestine – C188–189, C260), rectum (rectosigmoid junction – C199, rectum - C209), and small intestine (C170–173, C178-179). Grade of the tumor is categorized by SEER into 5 categories based on the level of differentiation (Grade 1 – well-differentiated, Grade 2 – moderately differentiated, Grade 3 – poorly differentiated, Grade 4 – undifferentiated, and Unknown). Surgical intervention is categorized into 4 factors, Local – local resection, Oncologic – radical resection, None – no operative intervention, and Unknown. Excluded patients were those with survival listed as zero months regardless of vital status, and those whose cause of death classification was N/A.

2.2. Statistical analysis

Demographic data and prognostic factors were compared based on location of the tumor. All categorical variables were compared using the Chi-squared test at 95% level of significance. Continuous variables such as follow-up time were compared using Kruskal Wallis ANOVA test for variance. Survival comparisons were conducted using Kaplan-Meier estimates. The multivariate Cox proportional-hazards model was used to compare survival between each tumor type and primary site with a 95% confidence interval (CI). An alpha of 5% with a *p*-value <0.05, and power of at least 80% was considered statistically significant. Post-hoc power calculation using the well-differentiated NETs of the colon sample size of n = 6783, and a median survival of 34 months, revealed a power of 100%, based on previously published SEER data.[4]

3. Results

46,665 patients were diagnosed with NENs of the colon (n = 10,518, 22.5%), rectum (18,063, 38.7%), and SI (18,084, 38.8%) from 2000 to 2017 (Table 1). Patients with small intestinal NENs were older (age 60–69 years, n = 5209, 29.3% of SI NENs) than patients with NENs of the colon (age <49 years, n = 3192, 30.9% of colon NENs, p < 0.001) or rectum (most common age 50–59 years, n = 7710, 43.6% of rectal NENs). Patients with NENs of the colon were predominantly female (n = 5722, 54.4%, p < 0.001), whereas those with small intestinal NENs were predominantly male (n = 9437, 52.2%), and nearly equivocal for rectal NENs (male n = 9074, 50.2%). Patients of minority race were predominantly diagnosed with rectal NENs (American Indian/Alaskan native n = 165, 53.3%; Asian n = 2454, 73%; Black n = 4315, 49.4%, p < 0.001), while White patients were more frequently diagnosed with NENs of the SI (n = 14284, 43.4%).

The majority of NENs diagnosed of the colon, rectum and SI were

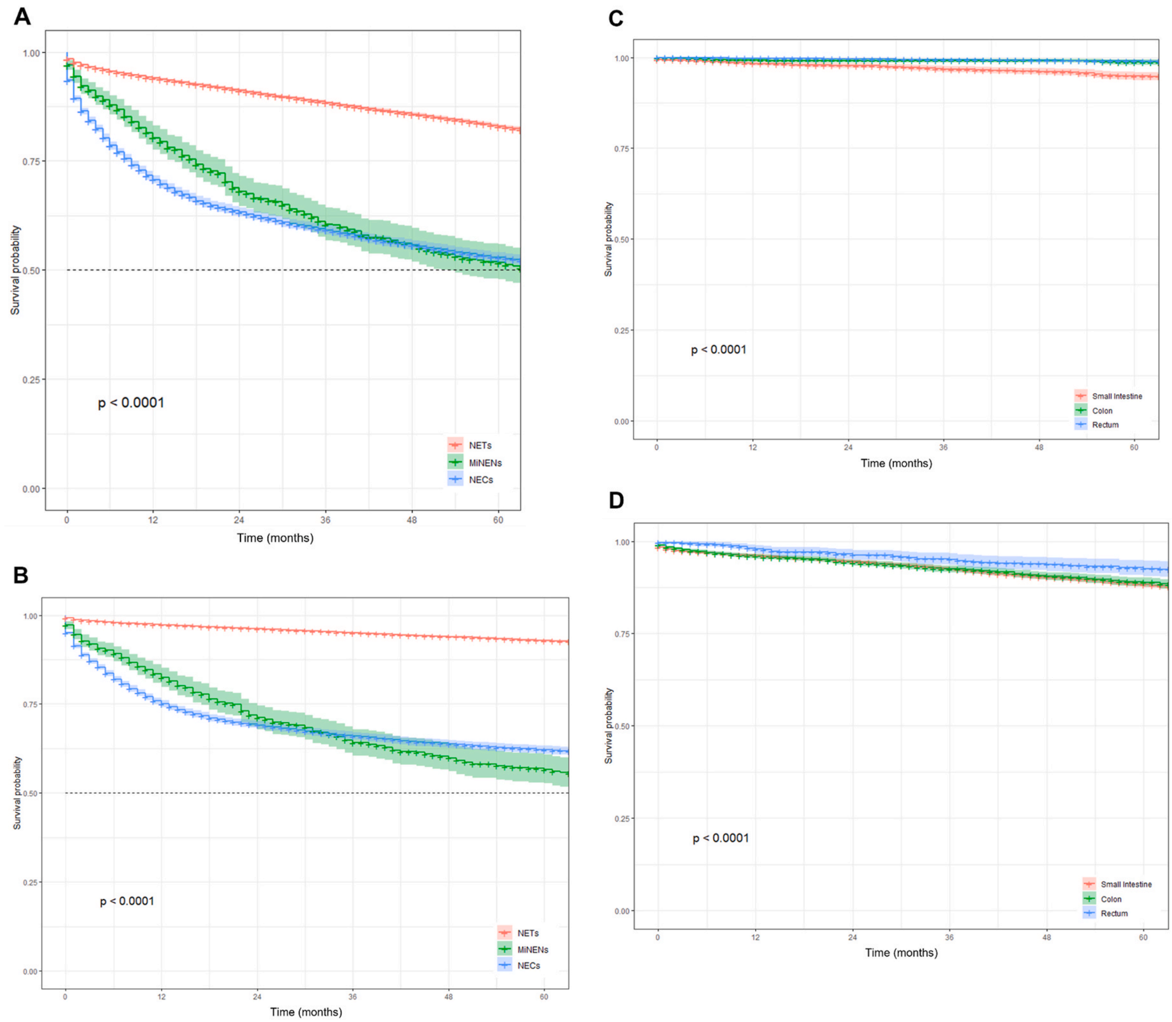

Fig. 2. Kaplan-Meier 5-year overall survival curves (**A**) and 5-year cancer-specific survival curves (**B**) by WHO histopathological classification: NETs (red), MiNENs (green), and NECs (blue). Kaplan-Meier 5-year cancer-specific survival curves by tumor site among patients with well-differentiated NETs who underwent local resection only (**C**), NETs upon oncologic resection only (**D**) in those with small intestinal (red), colonic (green), and rectal (blue) NETs. Patient data collected using the SEER 18 registry from 2000 to 2017. Abbreviations: Surveillance, Epidemiology, and End Results (SEER) database, World Health Organization (WHO), neuroendocrine neoplasms (NENs), well-differentiated neuroendocrine tumors (NETs), poorly differentiated neuroendocrine carcinomas (NECs), mixed neuroendocrine non-neuroendocrine neoplasms (MiNENs). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

well-differentiated NETs (n = 6783, 64.5% of colon; n = 15828, 87.6% of rectum; n = 14705, 81.3% of SI). Median survival for NENs of the rectum was longer (67 months, $p < 0.001$) than in the SI (52 months) and shortest in the colon (27 months) (Table 1). Similarly, 5-year OS and cancer-specific survival were better in rectal tumors (OS 87.3%, CSS 93.5%) compared to small intestine (OS 72.4%, CSS 86.5%), and colon (OS 65.5%, CSS 73.6%) (Fig. 1). On multivariate analysis, rectal site of origin was identified as an independent predictor of improved 5-year OS (OS HR 0.72, 95% CI 0.68–0.77, $p < 0.001$) but not cancer-specific survival (HR 1.16, 95% CI 1.07–1.27, $p < 0.001$) (Table 2). The variables that were independently prognostic of impaired OS and cancer-specific survival included older age (age >80 years), male sex, distant disease and NECs (Table 2).

Well-differentiated NETs were diagnosed more frequently in the rectum (n = 15828, 42.4%) (Table 1), and associated with improved 5-year OS and cancer-specific survival (OS 92.1%, CSS 98.1%) compared to those in the colon (OS 81.4%, CSS 89.7%) or SI (OS 73.5%, CSS 88.1%) (Fig. 2A and B). Similarly, most rectal tumors underwent local resection (n = 12949, 71.7%) (Table 1) and were associated with improved 5-year OS and cancer-specific survival (OS 93.3%, CSS 99%) compared to colon (OS 91.4%, CSS 98.7%) and SI (OS 78.3%, CSS 94.9%) well-differentiated NETs that underwent local resection (Fig. 2C).

Mixed neuroendocrine non-neuroendocrine neoplasms (MiNENs) were rare and more commonly located in the colon (n = 750, 86.6%; rectum n = 68, 0.08%; SI n = 48, 0.06%) (Table 1). Median survival for MiNENs of the colon was shorter (22 months, $p = 0.004$) than in the SI (31.5 months) and longest in the rectum (41 months) (Table 1). MiNENs of the SI had worse 5-year OS (49.4%) compared with colon (50.3%) and rectum (67.6%). Cancer-specific survival of SI MiNENs was increased (57.1%) compared to that of the colon (55.4%) but reduced compared to rectum (67.6%). Similarly, MiNENs of the SI who underwent oncologic resection were also associated with reduced 5-year OS (48.8%) and cancer-specific survival (52.1%) as compared to colon (OS 52.5%, CSS 57.9%) or rectum (CSS 60.7%).

The majority of poorly differentiated NECs were from the SI (n = 3331, 39.3% of NECs) (Table 1) and associated with a better 5-year OS (68.2%) compared with NECs of the rectum (52.2%, HR 1.54, 95% CI 1.38–1.71, $p < 0.001$) or colon (36.0%, HR 1.90, 95% CI 1.74–2.06, $p < 0.001$) (Table 3, Fig. 3). Median survival was longest for NECs of the SI (52 months, $p < 0.001$) than rectum (21 months), and shortest in the colon (11 months) (Table 1). There was no statistically significant

Table 3
Univariate and multivariate hazard ratio survival analysis of tumor types.

	Overall Survival				Cancer-specific Survival			
	Univariate HR (95% CI)		Multivariate HR (95% CI)		Univariate HR (95% CI)		Multivariate HR (95% CI)	
Well-differentiated NETs								
Age, years								
<49	–		–		–		–	
50–59	1.73 (1.56–1.91)	p < 0.001	1.76 (1.59–1.95)	p < 0.001	1.46 (1.25–1.72)	p < 0.001	1.49 (1.27–1.75)	p < 0.001
60–69	3.94 (3.57–4.35)	p < 0.001	3.53 (3.20–3.89)	p < 0.001	3.18 (2.74–3.70)	p < 0.001	2.54 (2.19–2.96)	p < 0.001
70–79	8.23 (7.46–9.07)	p < 0.001	6.77 (6.13–7.47)	p < 0.001	5.72 (4.92–6.65)	p < 0.001	4.05 (3.48–4.71)	p < 0.001
>80	18.36 (16.59–20.32)	p < 0.001	14.03 (12.66–15.56)	p < 0.001	11.27 (9.61–13.22)	p < 0.001	7.23 (6.15–8.50)	p < 0.001
Female	–		–		–		–	
Male	1.20 (1.15–1.25)	p < 0.001	1.21 (1.16–1.26)	p < 0.001	1.10 (1.02–1.18)	p = 0.010	1.11 (1.03–1.19)	p = 0.007
Tumor site								
Small Intestine	–		–		–		–	
Colon	0.63 (0.60–0.67)	p < 0.001	0.96 (0.90–1.02)	p = 0.161	0.78 (0.71–0.86)	p < 0.001	1.27 (1.15–1.39)	p < 0.001
Rectum	0.29 (0.27–0.30)	p < 0.001	0.56 (0.52–0.60)	p < 0.001	0.14 (0.13–0.16)	p < 0.001	0.66 (0.57–0.76)	p < 0.001
Stage								
Localized	–		–		–		–	
Regional	1.90 (1.80–2.01)	p < 0.001	1.16 (1.09–1.25)	p < 0.001	4.89 (4.40–5.44)	p < 0.001	2.74 (2.41–3.12)	p < 0.001
Distant	4.04 (3.82–4.28)	p < 0.001	2.37 (2.22–2.53)	p < 0.001	17.94 (16.28–19.76)	p < 0.001	9.18 (8.15–10.33)	p < 0.001
Unknown	1.31 (1.21–1.42)	p < 0.001	0.94 (0.86–1.02)	p = 0.132	2.06 (1.75–2.43)	p < 0.001	1.22 (1.02–1.45)	p = 0.030
Extent of resection								
Oncologic	–		–		–		–	
Local	0.40 (0.38–0.42)	p < 0.001	0.94 (0.88–1.01)	p = 0.112	0.13 (0.11–0.14)	p < 0.001	0.59 (0.51–0.70)	p < 0.001
None	1.07 (1.01–1.13)	p = 0.021	1.52 (1.42–1.63)	p < 0.001	0.92 (0.84–1.00)	p = 0.049	1.63 (1.47–1.80)	p < 0.001
Unknown	0.71 (0.56–0.89)	p = 0.004	1.43 (1.12–1.82)	p = 0.004	0.42 (0.27–0.66)	p < 0.001	1.44 (0.91–2.30)	p = 0.121
Poorly differentiated NECs								
Age, years								
<49	–		–		–		–	
50–59	1.18 (1.04–1.33)	p = 0.011	1.21 (1.06–1.37)	p = 0.003	1.10 (0.96–1.26)	p = 0.167	1.13 (0.98–1.29)	p = 0.084
60–69	1.70 (1.51–1.91)	p < 0.001	1.54 (1.36–1.73)	p < 0.001	1.43 (1.26–1.64)	p < 0.001	1.25 (1.09–1.43)	p = 0.001
70–79	2.77 (2.46–3.12)	p < 0.001	2.30 (2.04–2.59)	p < 0.001	2.18 (1.91–2.49)	p < 0.001	1.73 (1.51–1.97)	p < 0.001
>80	4.65 (4.11–5.25)	p < 0.001	3.62 (2.19–4.09)	p < 0.001	3.53 (3.08–4.05)	p < 0.001	2.63 (2.29–3.02)	p < 0.001
Female	–		–		–		–	
Male	1.15 (1.08–1.23)	p < 0.001	1.22 (1.14–1.29)	p < 0.001	1.15 (1.07–1.24)	p < 0.001	1.19 (1.11–1.28)	p < 0.001
Tumor site								
Small Intestine	–		–		–		–	
Colon	2.71 (2.52–2.92)	p < 0.001	1.90 (1.74–2.06)	p < 0.001	3.80 (3.47–4.16)	p < 0.001	2.47 (2.23–2.74)	p < 0.001
Rectum	1.60 (1.47–1.74)	p < 0.001	1.54 (1.38–1.71)	p < 0.001	2.21 (1.99–2.45)	p < 0.001	2.10 (1.86–2.38)	p < 0.001
Stage								
Localized	–		–		–		–	
Regional	2.08 (1.86–2.32)	p < 0.001	1.52 (1.35–1.72)	p < 0.001	4.05 (3.41–4.81)	p < 0.001	2.81 (2.34–3.39)	p < 0.001
Distant	6.25 (5.66–6.89)	p < 0.001	3.86 (3.44–4.32)	p < 0.001	15.51 (13.22–18.20)	p < 0.001	8.79 (7.37–10.48)	p < 0.001
Unknown	2.71 (2.28–3.23)	p < 0.001	1.71 (1.42–2.06)	p < 0.001	4.53 (3.53–5.81)	p < 0.001	2.70 (2.08–3.51)	p < 0.001
Extent of resection								
Oncologic	–		–		–		–	
Local	0.34 (0.30–0.39)	p < 0.001	0.83 (0.71–0.97)	p = 0.021	0.20 (0.16–0.25)	p < 0.001	0.68 (0.54–0.85)	p = 0.001
None	2.54 (2.38–2.71)	p < 0.001	1.70 (1.57–1.85)	p < 0.001	2.81 (2.61–3.02)	p < 0.001	1.58 (1.44–1.74)	p < 0.001
Unknown	1.08 (0.72–1.62)	p = 0.697	1.73 (1.14–2.63)	p = 0.010	0.86 (0.51–1.46)	p = 0.575	1.54 (0.89–2.66)	p = 0.120
MiNENs								
Age, years								
<49	–		–		–		–	
50–59	2.04 (1.39–2.99)	p < 0.001	2.04 (1.38–3.02)	p < 0.001	1.68 (1.13–2.52)	p = 0.011	1.55 (1.02–2.33)	p = 0.038
60–69	2.23 (1.52–3.26)	p < 0.001	2.31 (1.57–3.42)	p < 0.001	2.03 (1.37–3.01)	p < 0.001	1.94 (1.30–2.91)	p = 0.001
70–79	2.85 (1.91–4.26)	p < 0.001	3.22 (2.12–4.89)	p < 0.001	2.11 (1.37–3.25)	p = 0.001	2.19 (1.39–3.43)	p = 0.001
>80	5.17 (3.38–7.92)	p < 0.001	6.31 (4.07–9.78)	p < 0.001	3.93 (2.49–6.20)	p < 0.001	4.56 (2.86–7.28)	p < 0.001
Female	–		–		–		–	
Male	0.84 (0.68–1.03)	p = 0.090	0.96 (0.78–1.18)	p = 0.691	0.89 (0.71–1.11)	p = 0.316	1.04 (0.82–1.31)	p = 0.747
Tumor site								
Small Intestine	–		–		–		–	
Colon	0.83 (0.58–1.21)	p = 0.340	1.07 (0.71–1.61)	p = 0.732	1.00 (0.64–1.56)	p = 0.998	1.30 (0.80–2.11)	p = 0.297
Rectum	0.49 (0.29–0.84)	p = 0.009	0.71 (0.39–1.26)	p = 0.241	0.60 (0.32–1.11)	p = 0.105	0.98 (0.50–1.91)	p = 0.944
Stage								
Localized	–		–		–		–	
Regional	3.12 (2.14–4.55)	p < 0.001	2.71 (1.82–4.05)	p < 0.001	5.28 (3.11–8.95)	p < 0.001	4.53 (2.60–7.88)	p < 0.001
Distant	11.67 (8.08–16.85)	p < 0.001	9.61 (6.47–14.26)	p < 0.001	22.11 (13.18–37.09)	p < 0.001	17.97 (10.40–31.03)	p < 0.001
Unknown	4.28 (1.80–10.17)	p = 0.001	2.72 (1.08–6.83)	p = 0.034	6.23 (2.08–18.65)	p = 0.001	4.56 (1.44–14.40)	p = 0.010
Extent of resection								
Oncologic	–		–		–		–	
Local	0.37 (0.20–0.68)	p = 0.001	1.31 (0.65–2.66)	p = 0.450	0.26 (0.11–0.58)	p = 0.001	1.11 (0.42–2.90)	p = 0.835
None	3.49 (2.57–4.72)	p < 0.001	3.40 (2.37–4.87)	p < 0.001	3.67 (2.66–5.06)	p < 0.001	3.37 (2.29–4.96)	p < 0.001

WHO: World Health Organization. NETs: well-differentiated neuroendocrine tumors, NECs: poorly differentiated neuroendocrine carcinomas, MiNENs: mixed neuroendocrine non-neuroendocrine neoplasms. Surgery type none: not resected.

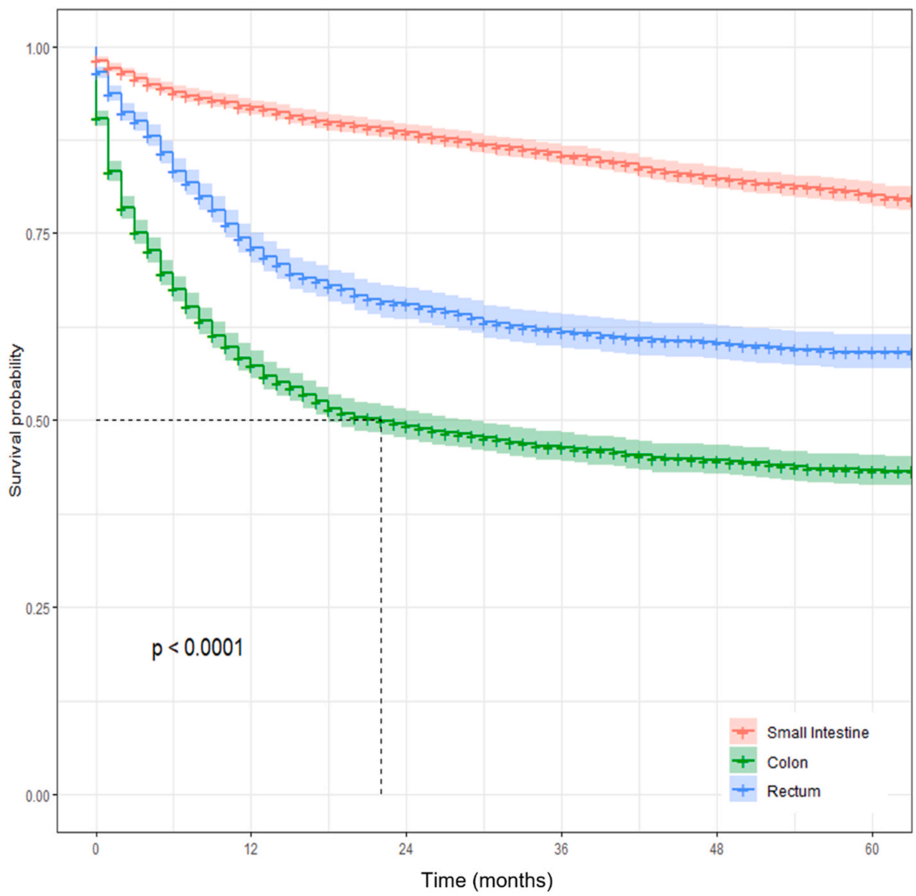

Fig. 3. Kaplan-Meier 5-year cancer-specific survival curves by tumor site among all poorly differentiated NECs. Patient data collected using the SEER 18 registry from 2000 to 2017 among patients diagnosed with small intestinal (red), colonic (green), and rectal (blue) NECs. Abbreviations: Surveillance, Epidemiology, and End Results (SEER) database, neuroendocrine neoplasms (NENs), well-differentiated neuroendocrine tumors (NETs), poorly differentiated neuroendocrine carcinomas (NECs). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

difference observed among patients with NECs who underwent local resection only (Fig. 4A). Among those with NECs who underwent oncologic resection, SI primary tumors were associated with better 5-year OS (72.5%) and cancer-specific survival (83.6%) compared to colon (OS 41.9%, CSS 49.8%), and rectum (OS 35.1%, CSS 39.4%) where survival was reduced (Fig. 4B).

While the sample size is large enough to detect small but statistically significant differences between age, gender, race and tumor site, the clinical significance of these differences is as yet unclear. However, on multivariate analysis, the relationships between variables may offer clinically relevant prognostic information. The narrow confidence intervals suggest the differences are statistically relevant. The clinical relevance is apparent in the sequential associated reduction of OS and CSS with increasing age, grade, differentiation, stage, and extent of resection (Table 2). These associations are expected but such a large sample size allows the small differences statistical significance. Similar to a previously published study, colonic NENs exhibited an associated reduced OS and CSS compared to rectal and SI NENs on multivariate analysis (Table 2).[4] This was consistent between poorly-differentiated NECs, however, well-differentiated NETs of the SI were associated with worse OS than colon and rectum (Table 3). Subpopulations within the group of MiNENs were too small to achieve statistical significance as this diagnosis is rare and comprises a heterogeneous group of pathologic characteristics that was unique to the updated WHO classification (Table 3).

4. Discussion

This study, using the SEER national database, demonstrates that those diagnosed with rectal NENs are younger with prolonged mean survival and improved 5-year overall and cancer-specific survival compared with SI NENs. Interestingly, the age-adjusted incidence rate of all rectal NENs was comparable to that of the SI, but that of the colon was significantly higher than both, likely due to increased screening with colonoscopy.[12,14–16,26,30] The diagnosis of SI NENs are increasingly found incidentally during diagnostic imaging or endoscopy, more so than gastric outlet obstruction or biliary obstruction symptoms with duodenal carcinoids.[26,30] While the majority of those diagnosed with NENs were histologically well-differentiated, regardless of site of origin, most of the well-differentiated NETs were of rectal origin, which likely contributed to their overall improved and prolonged survival. Unlike the colon and SI, most of those with rectal NETs underwent local resection rather than oncologic surgical resection, suggesting that rectal NETs were found at an earlier stage amenable to local resection. Unfortunately, whether these patients were diagnosed incidentally upon screening, or evaluated due to the presence of symptoms remains unclear due to the limitations of the database.

When comparing rectal with colonic NENs, both of which can be presumed to be found during screening colonoscopy, well-differentiated NETs of the colon had impaired 5-year OS and cancer-specific survival compared with that of the rectum. Colonic NENs in sum were associated with the shortest median survival and worst 5-year overall survival. Furthermore, poorly differentiated NECs of the colon had the worst 5-year OS compared with SI and rectum, implying that either colonic NECs are diagnosed at a later stage or the disease is more rapidly progressive. This argument is thus more complicated with the finding that among those patients with NECs who underwent oncologic surgical resection, those originating at the rectum instead had reduced 5-year overall survival compared with both colon and SI. This is limited by unknown margin status of patients, and the specific type and extent of surgical resection per case are not clearly stratified or subclassified. Despite these limitations, the data suggests that poorly differentiated

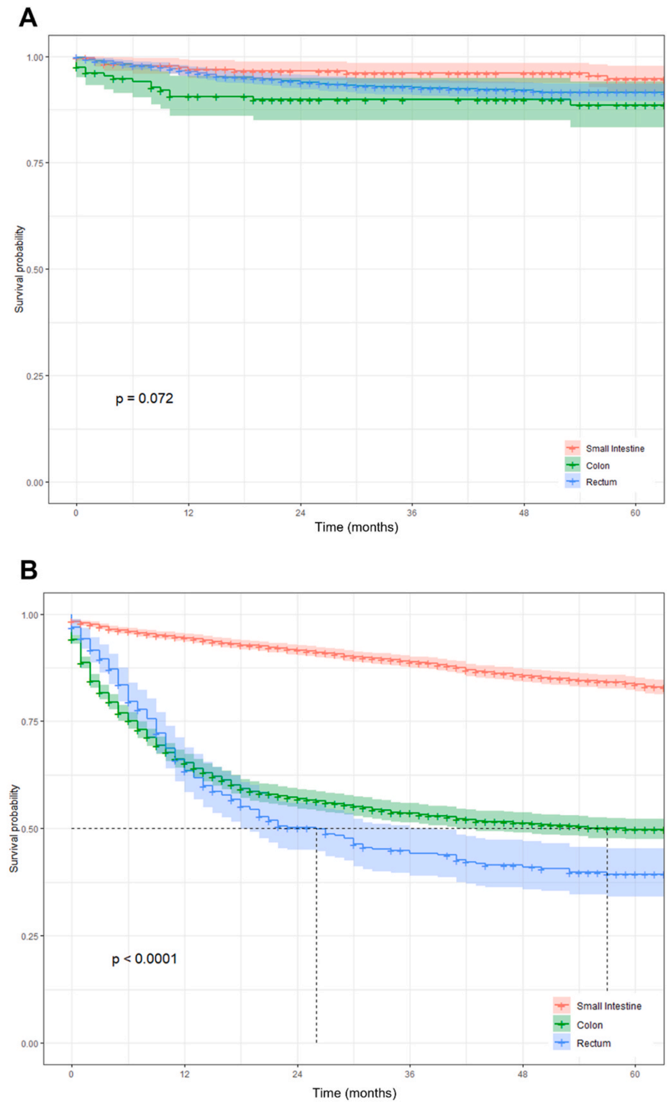

Fig. 4. Kaplan-Meier 5-year cancer-specific survival curves by tumor site among patients with poorly differentiated NECs after local resection only (**A**), and patients with poorly differentiated NECs who underwent oncologic surgical resection only (**B**). Patient data collected using the SEER 18 registry from 2000 to 2017 among patients diagnosed with small intestinal (red), colonic (green), and rectal (blue) NECs. Abbreviations: Surveillance, Epidemiology, and End Results (SEER) database, neuroendocrine neoplasms (NENs), well-differentiated neuroendocrine tumors (NETs), poorly differentiated neuroendocrine carcinomas (NECs). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

rectal NECs of patients who undergo extensive oncologic resection are thus more aggressive than that of colon or SI, but regardless of histological subtype, rectal NENs have a better prognosis than colon and SI.[33]

Recent studies have suggested an associated decrease in overall survival between patients with rectal NENs who were observed in comparison with those who underwent local excision or radical resection.[34] Prognostic factors associated with worse outcomes specific to rectal NENs include tumor size, WHO grade, depth of invasion beyond the muscularis layer, and lymph node involvement.[10,23,33,35,36] However, the biologic mechanisms specific to rectal NECs driving the poorer prognosis compared to colon or SI remains unclear. The SEER analysis by McConnell in 2016 found no difference in overall survival among lymph node positive patients who underwent local excision or biopsy alone.[37] The data suggested that perhaps once lymph node involvement occurs that local control of the primary rectal tumor may not mitigate regional spread. Furthermore, tumor size greater than 1 cm is suggested to be predictive of lymph node involvement.[38–40] Another alternative is that the histological subtype of poorly differentiated NECs may affect prognosis. For example, a Dutch study suggested improved outcomes for large-cell subtypes of NECs.[41–43] However, whether this can be attributed across different primary sites, when comparing colon, rectum and SI, further complicates the elucidation of possible biological mechanisms.[26,30,44]

Finally, independent prognostic factors on multivariate analysis revealed NENs of rectal origin associated with improved prognosis, and distant disease predictive of a poorer prognosis. The association of prognosis on WHO histological classification was statistically significant for well-differentiated NETs associated with improved prognosis, and both MiNENs and poorly differentiated NECs predictive of poor prognosis (Table 2). However, the hazard ratios were not as elevated as with distant disease, a known prognostic factor.[16–23] This may be due to the very few cases diagnosed with MiNENs as compared to well-differentiated NETs and poorly differentiated NECs. Similarly, prior studies remain unclear as to whether MiNENs are associated with a poorer prognosis than NECs,[45] which may be due to the recent inclusion within the updated 2019 WHO classification.

The limitations of this study include the retrospective design and lack of insight on case specifics, such as symptoms at presentation or incidental diagnosis, resection margin status, type of surgical procedure performed, whether the procedure was emergent or electively scheduled, and any information on recurrence or progression despite therapy. These factors could provide further insight into the aggressiveness of disease impacting the reduced prognosis in colonic compared to rectal NENs.[29] Additionally, being a retrospective study, any confounding factors cannot be entirely excluded as potentially influencing the data analysis. As previously discussed, limitations due to the large sample size allow an observed statistical significance for very small differences in survival or demographic characteristics between subsets. While the statistical significance is validated with an alpha p-value kept at 5%, the clinical significance is more difficult to ascertain when small differences are detected between nearly every subset of data. However, other studies have corroborated the findings as clinically relevant among demographic groups, and the survival differences between colonic NENs compared with SI and rectal NENs.[4] Ongoing research evaluating newer somatostatin analogs as systemic therapy either alone or in combination with cytotoxic agents for the treatment of advanced NENs may provide an optimistic impact in improved outcomes.[26–28]

5. Conclusion

Rectal NENs overall have an associated improved 5-year OS and CSS compared with SI and colonic NENs, likely due to the predominance of well-differentiated NETs of the rectum at diagnosis, and that most rectal NENs are amenable for local resection. Surgical resection is associated with improved 5-year OS for poorly-differentiated NECs of the SI and colon, except in the rectum where survival was reduced. More frequent surveillance and timely initiation of systemic therapy, including somatostatin analogs and chemotherapy, should be considered for poorly-differentiated rectal NECs.

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Declaration of competing interest

The authors declare no disclosures or competing interests.

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